SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4: *Design network topologies — star, mesh, bus, and hybrid — using networking devices, transmission media, and cabling standards.* ***(BL2, BL6)***

Assessment Tool Weightage: 30%

Questions: 5 Total Marks: 25

Annexure A

Comparative Analysis

Code of Conduct:

I S. Karthikeyan / 192419048 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CSA07 — COMPUTER NETWORKS

CO4 Assessment Tool 3 — Comparative Analysis: Answers

CO4: Design network topologies — star, mesh, bus, and hybrid

Q1. Star vs. Bus Topology — Cost, Reliability, and Performance

Star and bus are two of the earliest wired LAN topologies, and they sit at opposite ends of a trade-off between upfront cost and operational robustness.

Comparison

Criterion	Star Topology	Bus Topology
Cost	Higher — needs a central switch/hub plus a dedicated cable run to every node	Lower — a single shared backbone cable with simple drop connections
Reliability	High — one node or cable failing does not affect any other node	Low — a break anywhere on the shared backbone can bring down the whole segment
Performance	Better and more consistent — a switch gives each device its own collision-free link	Degrades as more devices are added, since all devices share one transmission medium

Why the Difference Exists

A modern star network uses a switch that gives every connected device its own dedicated, full-duplex link, so devices don't have to compete for the medium the way they do in a shared bus. A bus network instead relies on every device sharing one physical cable using CSMA/CD-style contention — the more devices attached, the more often two devices try to transmit at the same time, causing collisions and forcing retransmissions, which is exactly why bus performance drops as the network grows while star's does not.

Real-World Example

Older 10BASE2/10BASE5 Ethernet networks used a bus topology built on shared coaxial cable — cheap to wire, but a single loose connector could take the entire segment offline, and performance dropped noticeably as more machines were added. Essentially every wired office LAN built since has instead used star wiring through a switch, precisely because the reliability and performance gains were judged worth the extra cabling and hardware cost.

Conclusion: *Star trades a higher upfront cost for significantly better reliability and performance, which is exactly why it displaced bus topology as the standard for wired LANs — bus topology today is mostly of historical and academic interest.*

Q2. Mesh vs. Star Topology — Fault Tolerance

Fault tolerance is where mesh and star diverge most sharply, because of how each topology structurally handles a device or link failure.

Comparison

Criterion	Mesh Topology	Star Topology
Path redundancy	Multiple independent paths exist between any	Only one path exists — through the central

Criterion	Mesh Topology	Star Topology
	two nodes	device
Effect of a link/node failure	Traffic reroutes automatically through an alternate path	Only the device on that specific failed link loses connectivity
Effect of a central-point failure	No single central point exists to fail	The central hub/switch failing takes the entire network down
Overall fault tolerance	Very high	Moderate — bounded by the health of one device

Why the Difference Exists

In a full mesh of n nodes there are $n(n-1)/2$ independent links, so even several simultaneous link failures usually still leave at least one working path between any two nodes. A star network, by design, funnels every single connection through one central device — that device is a structural single point of failure that no amount of redundancy at the node level can work around.

Real-World Example

Internet backbone and ISP core networks are built as (partial) mesh topologies specifically so that a single fiber cut or router failure in one region does not take down connectivity for everyone behind it — traffic simply reroutes. A typical office LAN, in contrast, is star-wired through one switch; if that switch fails, the entire office loses network access until it is repaired or replaced.

Conclusion: *Mesh is the right choice whenever uptime is mission-critical (WAN cores, data-center backbones), while star's moderate fault tolerance — combined with its far lower cost and complexity — is perfectly adequate for ordinary LAN deployments where a brief switch outage is an inconvenience, not a catastrophe.*

Q3. Bus vs. Mesh Topology — Installation Complexity

Installation effort is one area where bus topology's simplicity stands in sharp contrast to mesh topology's demanding cabling requirements.

Comparison

Criterion	Bus Topology	Mesh Topology
Cabling required	One shared backbone cable plus a simple drop/tap per device	Every node connected to every other node — $n(n-1)/2$ links for a full mesh
Interfaces needed per node	1	$(n - 1)$ — one per every other node
How install effort scales	Roughly linear as devices are added (one more tap)	Grows quadratically as devices are added
Practical scale	Fine for small, low-cost networks	Only practical for a small number of critical nodes

Why the Difference Exists — Worked Example

The gap becomes obvious with real numbers. For 5 nodes, a bus needs just 1 backbone cable with 5 drop connections, while a full mesh needs $5\times4\div2 = 10$ separate links, with every node carrying 4 network interfaces. Scale up to 10 nodes and bus still needs only 1 backbone with 10 drops, while full mesh explodes to $10\times9\div2 = 45$

links, each node now needing 9 interfaces. This quadratic growth is exactly why full mesh is essentially never used to connect ordinary end-user devices.

Real-World Example

Mesh wiring in practice is reserved for a small handful of high-value nodes — for instance 4 to 8 core routers or data-center spine switches, where the cabling cost is justified by the fault tolerance gained. Bus topology, historically, was used for small departmental LANs precisely because it needed almost no planning or specialised hardware to install.

Conclusion: *Bus's installation simplicity made it attractive for small, cost-constrained networks in the past, while mesh's much higher cabling and hardware demands confine its practical use today to a small set of core nodes that specifically need maximum redundancy — not to general end-device connectivity.*

Q4. Hybrid vs. Star Topology — Scalability

Scalability is the clearest advantage hybrid topology holds over a pure star design, precisely because of how each one handles growth beyond its original size.

Comparison

Criterion	Hybrid Topology	Star Topology
Basic structure	Two or more topologies combined (e.g., several star segments joined by a backbone)	One central device with every node attached directly to it
How it expands	Add a new star segment or branch off the shared backbone	Add more ports to (or replace) the single central device
Growth ceiling	Very high — limited mainly by backbone capacity	Limited by the central device's port count
Typical scale of use	Large enterprises, campus networks with multiple departments/buildings	A single office, floor, or small LAN

Why the Difference Exists

A pure star network's growth is capped by definition: every device competes for a port on one central switch, and once that switch's ports are full, the only way to expand is to add a second switch and link it to the first — at which point the network has, structurally, already become a hybrid topology. Hybrid design simply embraces this from the start, letting each department or floor run its own star "cell," all tied into a larger shared backbone.

Real-World Example

A university or corporate campus network is the standard real-world example: each building runs its own star-wired LAN through local access switches, and every building's switch connects up into a shared core/distribution backbone (often itself partially meshed for redundancy). This is the industry-standard three-tier access–distribution–core architecture, and it is a hybrid topology by definition.

Conclusion: *Hybrid isn't really a competing alternative to star so much as what a star network naturally becomes once it needs to grow past a single switch's capacity — for any network expected to scale, designing it as a hybrid from the outset is the standard, scalable approach used in real enterprise networking.*

Q5. Maintenance Comparison — Star, Mesh, Bus, and Hybrid Topologies

Maintenance ease depends heavily on how easily a fault can be isolated to a specific point in the network, which varies a great deal across all four topologies.

Comparison

Topology	Fault Isolation	Troubleshooting Effort	Overall Maintenance
Star	Easy — traced to one node/cable via the central device's port status	Low	Easy
Mesh	Hard — many redundant links must be individually checked or monitored	High	Difficult
Bus	Hard — a break anywhere on the shared backbone is not obvious to locate	High	Difficult (harder than star)
Hybrid	Depends on the segment — local star cells are easy, the backbone less so	Medium	Moderate

Why the Difference Exists

A star network's centralized wiring closet means an administrator can see per-port link/activity status (or SNMP statistics) directly on the switch and pinpoint a failed port or cable almost immediately. A bus network offers no such central visibility — locating a break in the shared backbone typically requires physically tracing or testing the cable with a tester or time-domain reflectometer. Mesh is difficult to maintain for the opposite reason it is reliable to use: the same redundancy that provides multiple paths also means there are many more individual links to monitor, and identifying exactly which one has degraded requires proper network-monitoring tooling. Hybrid inherits a mix of both worlds — its local star segments are easy to diagnose, but the backbone connecting them needs broader monitoring and more skilled administration.

Real-World Example

Enterprise IT teams commonly rely on SNMP-based dashboards that work very well for star and hybrid access-layer segments, since each port reports its own status directly. Monitoring a meshed core, by contrast, typically needs more sophisticated tools — topology mapping and path-analysis utilities like traceroute — simply because there are far more possible paths and failure points to account for.

Conclusion: *Maintenance ease broadly tracks inversely with topology complexity — star is simplest to maintain, hybrid is moderate, and bus and mesh are both difficult, though for different reasons (bus from a lack of central visibility, mesh from sheer link count). Notably, mesh is still deliberately chosen despite being the hardest to maintain, wherever uptime matters more than administrative convenience — a clear illustration of the classic engineering trade-off between reliability and manageability.*